

Channel Estimation for mmWave MIMO using sub-6 GHz Out-of-Band Information

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T. Zemen and C. F. Mecklenbräuker

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- increasing demand for high data rate transmission

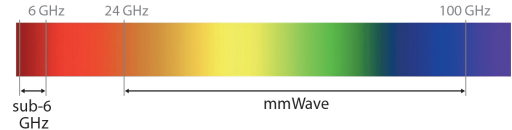
Introduction

- increasing demand for high data rate transmission
- sub-6 GHz band cannot meet these requirements



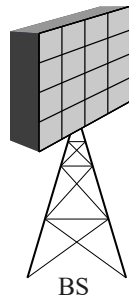
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- increasing demand for high data rate transmission
- sub-6 GHz band cannot meet these requirements
- mmWave bands allow for significantly higher data rates



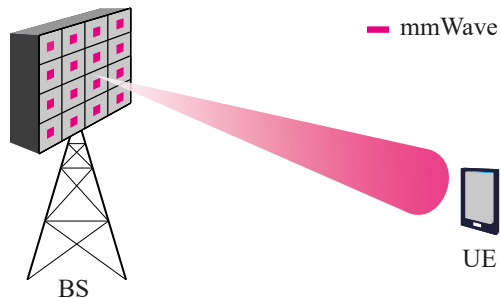
Motivation

- mmWave systems utilize antenna arrays



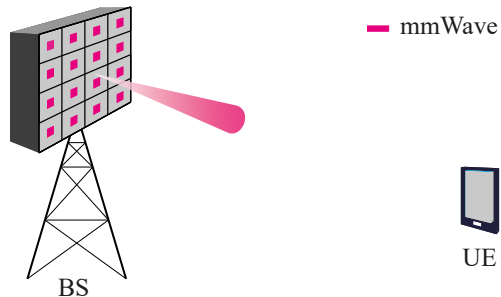
Motivation

- mmWave systems utilize antenna arrays
 - directional beamforming



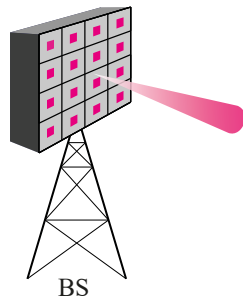
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 - **problem:** challenging link establishment

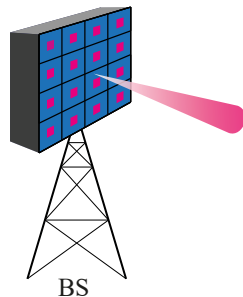


— mmWave



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- mmWave systems utilize antenna arrays
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 - **problem:** challenging link establishment
- mmWave deployment in conjunction with sub-6 GHz systems



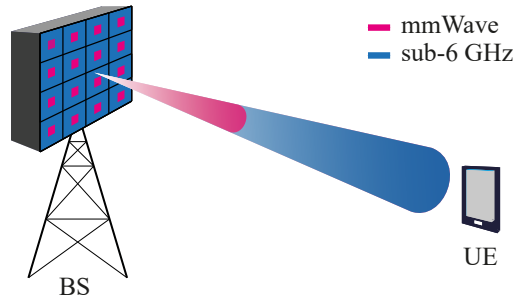
— mmWave
— sub-6 GHz



UE

Motivation

- mmWave systems utilize antenna arrays
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 - low pre-beamforming SNR
 - **problem:** challenging link establishment
- mmWave deployment in conjunction with sub-6 GHz systems
- use two bands cooperatively to achieve a better system throughput



- **three novel pilot-aided channel estimation methods proposed for mmWave MIMO systems**

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 - estimate mmWave channel with the aid of out-of-band sub-6 GHz information

- **three novel pilot-aided channel estimation methods proposed for mmWave MIMO systems**
 - estimate mmWave channel with the aid of out-of-band sub-6 GHz information
- **evaluate proposed channel estimation methods through simulations in terms of spectral efficiency**

System Model

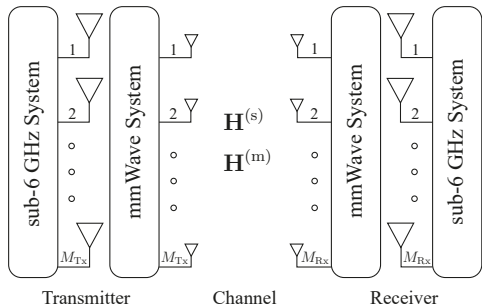
Channel Estimation Methods

Simulation-based Comparison

Summary and Conclusion

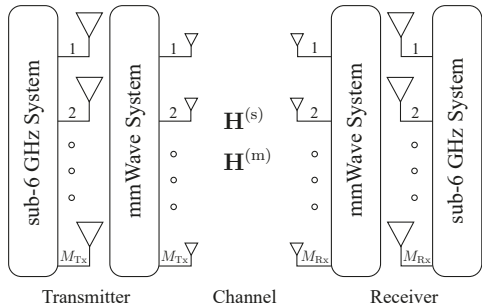
System Model

- multi-band MIMO OFDM system
 - co-located and aligned antenna arrays



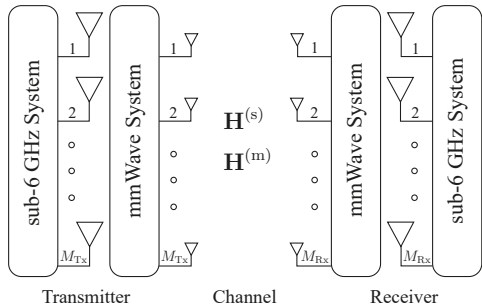
System Model

- multi-band MIMO OFDM system
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 - fully digital beamforming



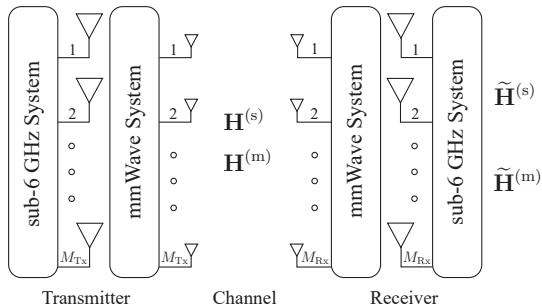
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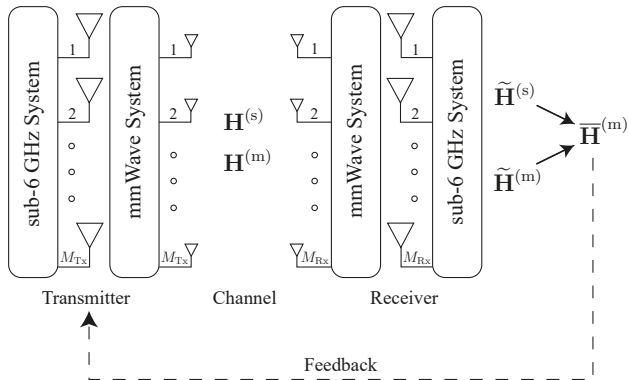
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 - training phase $\Rightarrow \tilde{\mathbf{H}}^{(s)}$ and $\tilde{\mathbf{H}}^{(m)}$



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 - feedback to the transmitter: $\bar{\mathbf{H}}^{(m)}$



System Model

- multi-band MIMO OFDM system
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 - training phase $\Rightarrow \tilde{\mathbf{H}}^{(s)}$ and $\tilde{\mathbf{H}}^{(m)}$
 - feedback to the transmitter: $\bar{\mathbf{H}}^{(m)}$
 - data transmission

$$\mathbf{y} = \bar{\mathbf{Q}}^H \mathbf{H}^{(m)} \bar{\mathbf{F}} \mathbf{x} + \bar{\mathbf{Q}}^H \mathbf{w}$$

receive symbol vector $\mathbf{y} \in \mathbb{C}^{\ell_{\max} \times 1}$

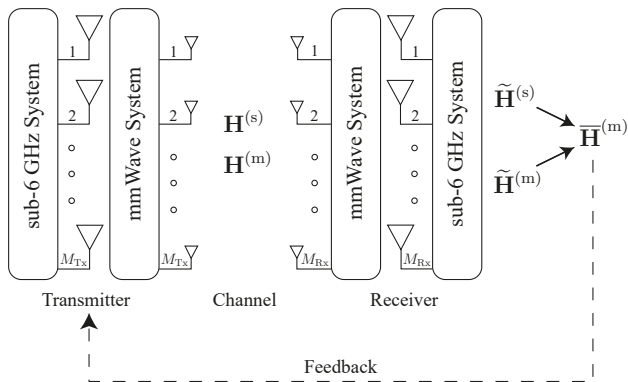
transmit symbol vector $\mathbf{x} \in \mathbb{C}^{M_{\text{Tx}} \times 1}$

precoding matrix $\bar{\mathbf{F}} \in \mathbb{C}^{M_{\text{Tx}} \times \ell_{\max}}$

combining matrix $\bar{\mathbf{Q}} \in \mathbb{C}^{M_{\text{Rx}} \times \ell_{\max}}$

additive white Gaussian noise

$\mathbf{w} \in \mathbb{C}^{M_{\text{Rx}} \times 1}$



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- sub-6 GHz system has higher SNR $\Rightarrow$ more accurate channel estimation

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- rotate the phase of the sub-6 GHz channel estimate

$$\begin{aligned} \tilde{\mathbf{H}}^{(s)} \odot e^{j2\pi \mathbf{D} \xi} &= \sqrt{\eta^{(s)}} \left(\sqrt{\frac{K^{(s)}}{1+K^{(s)}}} \tilde{\mathbf{H}}_{\text{fs}}^{(s)} + \sqrt{\frac{1}{1+K^{(s)}}} \tilde{\mathbf{H}}_{\text{rp}}^{(s)} \right) \odot e^{j2\pi \mathbf{D} \xi}, \quad \xi = \frac{1}{\lambda^{(s)}} - \frac{1}{\lambda^{(m)}} \\ &= \sqrt{\eta^{(s)}} \left(\sqrt{\frac{K^{(s)}}{1+K^{(s)}}} \underbrace{\tilde{\mathbf{H}}_{\text{fs}}^{(s)} \odot e^{j2\pi \mathbf{D} \xi}}_{\approx \tilde{\mathbf{H}}_{\text{fs}}^{(m)}} + \sqrt{\frac{1}{1+K^{(s)}}} \tilde{\mathbf{H}}_{\text{rp}}^{(s)} \odot e^{j2\pi \mathbf{D} \xi} \right) \end{aligned}$$

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Channel Estimation Methods

- translating
 - use the phase-rotated sub-6 GHz channel estimate

$$\overline{\mathbf{H}}^{(m)} = \tilde{\mathbf{H}}^{(s)} \odot e^{j2\pi \mathbf{D}\xi}$$

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- averaging
 - better results on average than the translating method

$$\overline{\mathbf{H}}^{(m)} = \frac{\tilde{\mathbf{H}}^{(s)} \odot e^{j2\pi \mathbf{D}\xi} + \tilde{\mathbf{H}}^{(m)}}{2}$$

Channel Estimation Methods

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$$\overline{\mathbf{H}}^{(m)} = \frac{\tilde{\mathbf{H}}^{(s)} \odot e^{j2\pi\mathbf{D}\xi} + \tilde{\mathbf{H}}^{(m)}}{2}$$

- weighting
 - weighting the sub-6 GHz and mmWave channel estimates by a factor W
 - select an optimum W for each specific combination of K -factor $K^{(m)}$ and SNR $\gamma^{(m)}$

$$\overline{\mathbf{H}}^{(m)} = W \left(K^{(m)}, \gamma^{(m)} \right) \tilde{\mathbf{H}}^{(s)} \odot e^{j2\pi\mathbf{D}\xi} + \left(1 - W \left(K^{(m)}, \gamma^{(m)} \right) \right) \tilde{\mathbf{H}}^{(m)}$$

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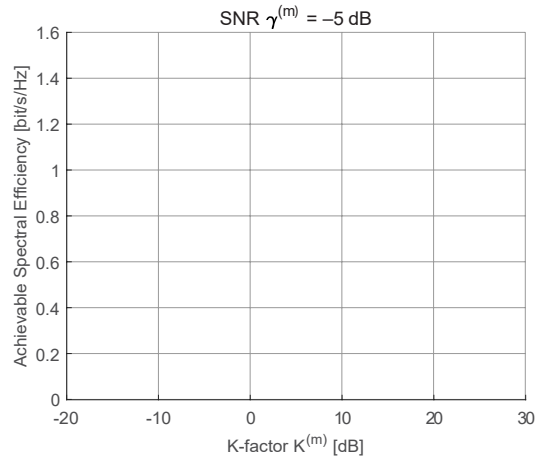
Simulation-based Comparison

- achievable spectral efficiency (SE)

Parameter	Value	
Frequency Band	sub-6 GHz	mmWave
Carrier Frequency f_c [GHz]	2.55	25.5
Wavelength λ [cm]	11.76	1.176
Bandwidth B [MHz]	10.08	100.8
Subcarrier Spacing Δf [kHz]	60	60
Cyclic Prefix t_{CP} [μ s]	1.19	1.19
Noise Figure F [dB]	3	3
Antenna Spacing [cm]	0.05λ	0.5λ
Channel Model	TDL-A	TDL-A

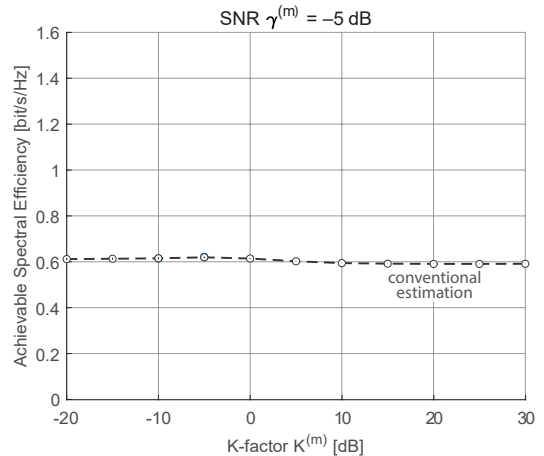
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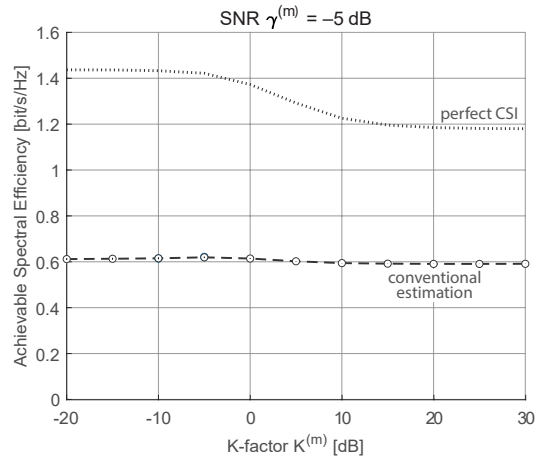
Simulation-based Comparison

- achievable spectral efficiency (SE)
- conventional channel estimation



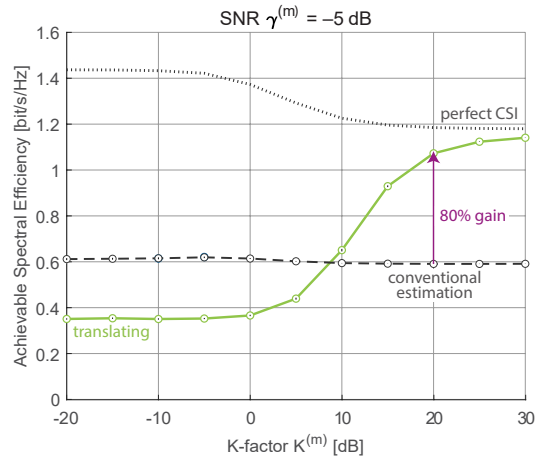
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- perfect CSI case



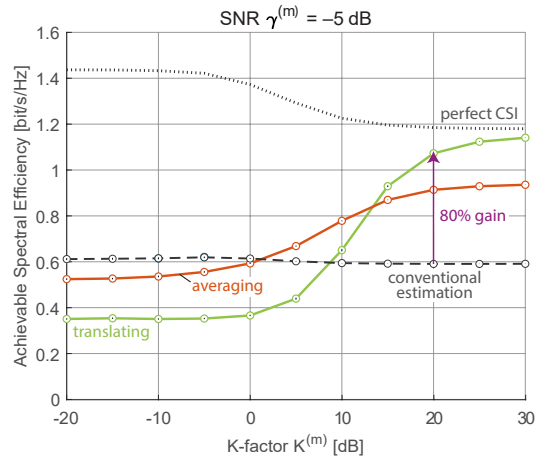
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- translating
 - up to 80% improvement at high K-factor
 - poor performance at low K-factor



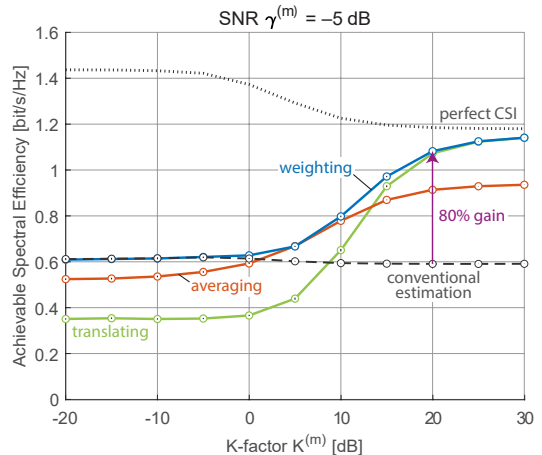
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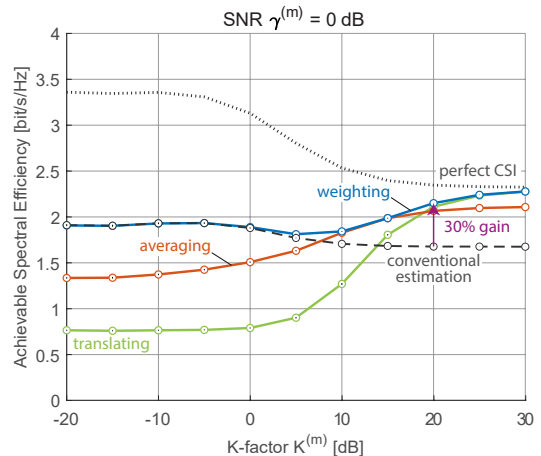
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- translating
 - up to 80% improvement at high K-factor
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- averaging
 - better performance at low K-factor
- weighting
 - improved performance for all K-factors



Simulation-based Comparison

- achievable spectral efficiency (SE)
- conventional channel estimation
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- translating
 - up to 80% improvement at high K-factor
 - poor performance at low K-factor
- averaging
 - better performance at low K-factor
- weighting
 - improved performance for all K-factors
- increasing SNR



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- significant increase in the achievable SE in the low-SNR regime

Summary and Conclusion

- SNR and K-factor affect the performance to a great extent
- significant increase in the achievable SE in the low-SNR regime
- negligible improvement in the high-SNR regime

Thank You !

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Channel Model

- channel model

$$\mathbf{H}^{(b)}[n] = \underbrace{\sqrt{\eta^{(b)}}}_{\text{pathloss}} \left(\underbrace{A_{\text{fs}}^{(b)} \mathbf{H}_{\text{fs}}^{(b)}[n]}_{\text{free-space}} + \underbrace{A_{\text{rp}}^{(b)} \mathbf{H}_{\text{rp}}^{(b)}[n]}_{\text{Rayleigh}} \right) = \underbrace{\sqrt{\eta^{(b)}}}_{\text{pathloss}} \left(\underbrace{\sqrt{\frac{K^{(b)}}{1+K^{(b)}}} \mathbf{H}_{\text{fs}}^{(b)}[n]}_{\text{free-space}} + \underbrace{\sqrt{\frac{1}{1+K^{(b)}}} \mathbf{H}_{\text{rp}}^{(b)}[n]}_{\text{Rayleigh}} \right)$$

$$\mathbf{H}_{\text{fs}}^{(b)}[n] = e^{-j \frac{2\pi}{\lambda^{(b)}} \mathbf{D}}, \quad \mathbf{D} \in \mathbb{R}^{M_{\text{Rx}} \times M_{\text{Tx}}}$$

- frequency-band dependent values

$$\text{Pathloss :} \quad \eta^{(m)} = \left(\frac{4\pi d f_c^{(m)}}{c_0} \right)^2 = \left(\frac{4\pi d \sqrt{\alpha} f_c^{(s)}}{c_0} \right)^2 = \alpha \eta^{(s)}, \quad \sqrt{\alpha} = \frac{f_c^{(m)}}{f_c^{(s)}}$$

$$\text{SNR :} \quad \gamma^{(m)} = \frac{P_{\text{T}}}{\eta^{(m)} k_{\text{B}} T B^{(m)}} = \frac{P_{\text{T}}}{\alpha \eta^{(s)} k_{\text{B}} T \beta B^{(s)}} = \gamma^{(s)} \frac{1}{\alpha \beta}, \quad \beta = \frac{B^{(m)} F^{(m)}}{B^{(s)} F^{(s)}}$$

Training Phase

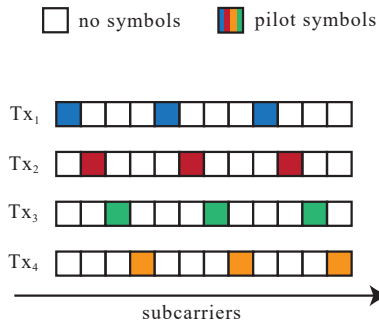
- training phase

$$\mathbf{y}^{(b)} = \mathbf{H}^{(b)} \boldsymbol{\phi}^{(b)} + \mathbf{w}^{(b)}$$

pilot symbols $\boldsymbol{\phi}^{(b)} \in \mathbb{C}^{M_{\text{Tx}} \times 1}$

channel matrix $\mathbf{H}^{(b)} \in \mathbb{C}^{M_{\text{Rx}} \times M_{\text{Tx}}}$

additive white Gaussian noise $\mathbf{w}^{(b)} \in \mathbb{C}^{M_{\text{Rx}} \times 1}$



Training Phase

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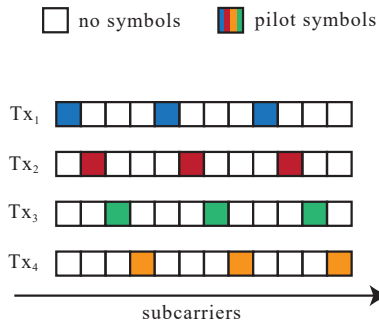
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- least square estimation and interpolation



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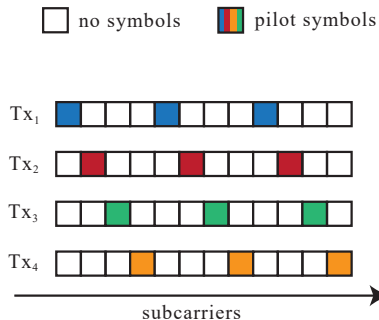
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- least square estimation and interpolation
- estimated channels $\tilde{\mathbf{H}}^{(s)}$ and $\tilde{\mathbf{H}}^{(m)}$



Training Phase

- training phase

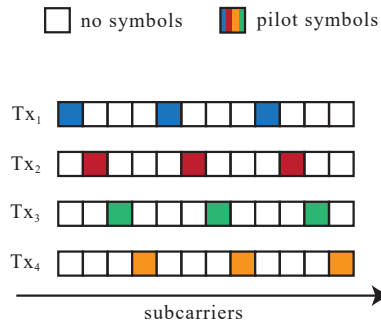
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- estimated channels $\tilde{\mathbf{H}}^{(s)}$ and $\tilde{\mathbf{H}}^{(m)}$
- feedback to the transmitter



System Model - Data Transmission

- only mmWave system considered

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- precoding matrix $\bar{\mathbf{F}}$ and combining matrix $\bar{\mathbf{Q}}$ obtained using SVD

$$\bar{\mathbf{H}}^{(m)} = \bar{\mathbf{Q}}\bar{\mathbf{\Sigma}} \left(\bar{\mathbf{F}}\mathbf{P}^{1/2} \right)^H$$

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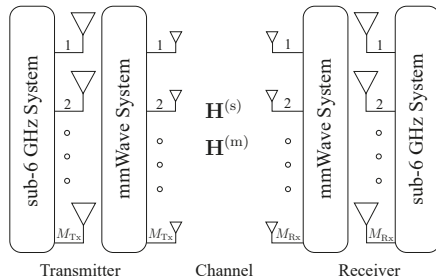
$$\mathbf{y} = \bar{\mathbf{Q}}^H \mathbf{H}^{(m)} \mathbf{P}^{1/2} \bar{\mathbf{F}} \mathbf{x} + \bar{\mathbf{Q}}^H \mathbf{w}$$

with receive symbol vector $\mathbf{y} \in \mathbb{C}^{\ell_{\max} \times 1}$, transmit symbol vector $\mathbf{x} \in \mathbb{C}^{M_{\text{Tx}} \times 1}$, channel matrix $\mathbf{H}^{(m)}$, power loading matrix $\mathbf{P}$ and additive white Gaussian noise $\mathbf{w}$

- power loading matrix $\mathbf{P}$ chosen such that

$$\left\| \bar{\mathbf{F}} \mathbf{P}^{1/2} \right\|_F^2 = P_T$$

- point-to-point multi-band MIMO system

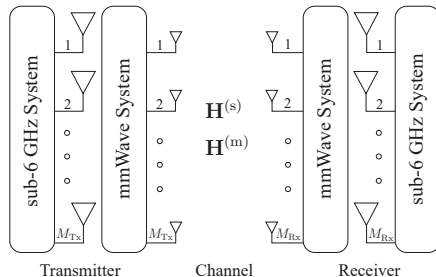


[1] B. Getu and J. Andersen, The MIMO cube - a compact MIMO antenna, IEEE Transactions on Wireless Communications, vol. 4, no. 3 pp. 11361141, 2005.

[2] R. Hansen and R. Collin, Smart Antenna Handbook, Wiley

Antenna Array Design

- point-to-point multi-band MIMO system
- co-located and aligned antenna arrays
 - uniform linear array (ULA) with the same number of elements
 - antenna spacing $\Rightarrow 0.5 \lambda^{(m)}$
 - compact design for sub-6 GHz array required [1]
 - smart antenna handbook

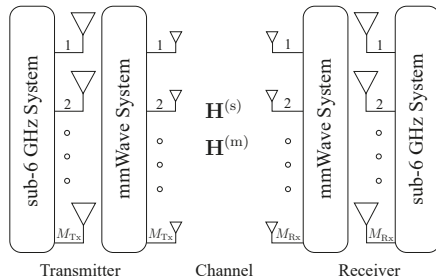


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- one RF chain per antenna $\Rightarrow$ fully digital beamforming



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Calculating Spectral Efficiency

- achievable spectral efficiency as the main performance metric for MIMO systems in bits/s/Hz

$$\text{SE} = \log_2 (1 + \text{SINR})$$

with

$$\text{SINR} = \frac{\sum_{\substack{\mu=1 \\ \nu=\mu}}^{\ell_{\max}} |\overline{G}_{\mu,\nu}|^2}{\sum_{\mu=1}^{\ell_{\max}} \sum_{\substack{\nu=1 \\ \nu \neq \mu}}^{\ell_{\max}} |\overline{G}_{\mu,\nu}|^2 + \sigma_w^2 \|\overline{\mathbf{Q}}\|_F^2}.$$

- $\overline{G}_{\mu,\nu}$ with $\mu, \nu \in \{1, \dots, \ell_{\max}\}$ represent entries of the channel gain matrix $\mathbf{G} \in \mathbb{C}^{\ell_{\max} \times \ell_{\max}}$ which is given by

$$\overline{\mathbf{G}} = \overline{\mathbf{Q}}^H \mathbf{H}^{(m)} \mathbf{P}^{1/2} \overline{\mathbf{F}}$$

Weighting Method - Training Phase

- weighting the sub-6 GHz and mmWave channel estimates by a weighting factor $W \in [0, 1]$

$$\bar{\mathbf{H}}^{(m)} = W \left(K^{(m)}, \gamma^{(m)} \right) \tilde{\mathbf{H}}^{(s)} \odot e^{j2\pi \mathbf{D} \xi} + \left(1 - W \left(K^{(m)}, \gamma^{(m)} \right) \right) \tilde{\mathbf{H}}^{(m)}$$

- training phase
 - determine and store $W \left(K^{(m)}, \gamma^{(m)} \right)$ in lookup table
 - assume perfect knowledge of the mmWave channel $\mathbf{H}^{(m)}$, the $\gamma^{(m)}$ and $K^{(m)}$
 - find W that results in the minimum mean channel estimation error

$$W \left(K^{(m)}, \gamma^{(m)} \right) = \underset{W}{\operatorname{argmin}} E \left\{ W \left(K^{(m)}, \gamma^{(m)} \right) \right\},$$

where the mean channel estimation error is defined by

$$E \left\{ W \left(K^{(m)}, \gamma^{(m)} \right) \right\} = \frac{1}{L} \sum_{l=1}^L \left\| \mathbf{H}_l^{(m)} - \bar{\mathbf{H}}_l^{(m)} \right\|_F^2.$$

actual channel coefficient $\mathbf{H}_l^{(m)}$, estimated channel coefficient for channel realization l $\bar{\mathbf{H}}_l^{(m)}$

Weighting Method - Data Transmission

- data transmission
 - previously calculated $W(K^{(m)}, \gamma^{(m)})$ taken from the lookup table
 - used to combine sub-6 GHz and mmWave channel estimates.

$K^{(m)}$ [dB]	$\gamma^{(m)}$ [dB]							
	-15	-10	-5	0	5	10	15	20
-20	0.83	0.92	0.98	1	1	1	1	1
-15	0.81	0.91	0.97	0.99	1	1	1	1
-10	0.76	0.86	0.95	0.99	1	1	1	1
-5	0.63	0.75	0.89	0.97	1	1	1	1
0	0.4	0.55	0.76	0.89	0.99	1	1	1
5	0.19	0.33	0.56	0.79	0.93	1	1	1
10	0.1	0.18	0.35	0.61	0.85	0.99	1	1
15	0.04	0.1	0.19	0.39	0.66	0.87	0.99	1
20	0.01	0.03	0.1	0.2	0.41	0.68	0.87	0.99
25	0	0	0.02	0.1	0.2	0.41	0.68	0.87
30	0	0	0	0.02	0.1	0.2	0.41	0.68